

KrishiSetu (SIH26132)

Full-Stack Technical Architecture, Process Flow & External Integrations Manual

1. Executive Summary & Architecture Overview

KrishiSetu is an agricultural market intelligence and direct farmer-buyer linkage platform engineered for Smart India Hackathon (SIH). It eliminates middleman exploitation, calculates true post-freight net profit margins across APMC Mandis, provides AI-driven price trajectory forecasting, and facilitates direct contracting with verified institutional buyers.

The platform is structured into three integrated tiers:

- **Frontend Client (React 19 + Vite + Tailwind CSS v4):** Role-specific portals (Farmer, Buyer, Mandi Officer), real-time browser GPS geolocation, interactive Recharts price visualizations, embedded route maps, and zero-config file persistence.
- **Backend Core (Java 17 + Spring Boot 3.3):** Enterprise REST API with Spring Security (stateless JWT), Spring Data JPA, PostgreSQL / H2 persistence, Flyway schema migrations, and recommendation algorithms.
- **AI Prediction Engine (Python FastAPI + Scikit-Learn):** Machine learning microservice utilizing RandomForestRegressor with sinusoidal seasonality curves and volume elasticity models to predict 1-day, 3-day, and 7-day commodity prices.

2. All External APIs, Websites & Third-Party Services Used

The table below details all external web services, public APIs, geocoding endpoints, and cloud CDNs integrated into the platform:

External Resource	Endpoint / Web Link	Technical Purpose & Functionality	Referenced File(s)
OpenStreetMap Nominatim API	https://nominatim.openstreetmap.org/reverse?format=json&lat={lat}&lon={lon}&zoom=10&addressdetails=1	Reverse Geocoding: Converts raw browser GPS latitude/longitude into human-readable Indian addresses (Village, District, State).	locationService.js
Google Maps Live Navigation	https://www.google.com/maps/dir/?api=1&origin={origin}&destination={lat},{lng}&travelmode=driving	1-Click Turn-by-Turn Driving Navigation: Generates mobile navigation intent launching Google Maps for haulage trucks directly to APMC gate.	LiveRouteMap.jsx
Google Maps Dynamic Embed	https://maps.google.com/maps?saddr={orig_lat},{orig_lng}&daddr={dest_lat},{dest_lng}&output=embed	Embedded Interactive Route Map: Displays transit corridors, distance, and highway routes directly inside the farmer dashboard.	LiveRouteMap.jsx
Data.gov.in / Agmarknet Mandi API	https://api.data.gov.in/resource/9ef84268-d588-465a-a308-a864a43d0070	Official Govt Mandi Feed: Daily arrival volumes, minimum, maximum, and modal wholesale prices across Indian APMC yards.	MandiApiClient.java, application.yml
Unsplash High-Res CDN	https://images.unsplash.com/photo-1595855759920...	Mandi Photography CDN: High-resolution market yard imagery mapped by APMC hub (Lasalgaon, Azadpur, Vashi, Gultekdi, etc.).	mandiImageService.js
FastAPI AI Service REST Endpoint	http://localhost:8000/predict	Microservice Communication: Spring 6 RestClient calling Python FastAPI RandomForest regression model.	AIServiceClient.java, main.py
Springdoc OpenAPI / Swagger UI	http://localhost:8080/swagger-ui.html & /v3/api-docs	API Documentation: Interactive OpenAPI 3.0 testing suite and automated schema explorer.	OpenApiConfig.java, application.yml

3. Complete File-by-File Architecture Breakdown

A. Python AI Price Prediction Microservice (ai-service/)

- **app/main.py:** FastAPI server initialization, CORS policy setup, health probes, and prediction endpoint routing.
- **app/model.py:** Houses AgriPriceModel. Trains RandomForestRegressor on 3-year simulated seasonal patterns (day-of-year sine cycles, arrival volume elasticity). Incorporates crop multipliers (Onion: +0.7%/day, Tomato: +0.9%/day) and sentiment factors (Monsoon +3.2%, Inflow volume +2.5%, Logistics -0.8%).
- **app/schemas.py:** Pydantic schemas validating request/response DTOs (PredictionRequest, PredictionResponse, PredictionPoint, FactorImpact).
- **requirements.txt & test_api.py:** Microservice dependencies (FastAPI, Scikit-Learn, XGBoost, Pandas, Numpy) and test suite.

B. Java Spring Boot 3.3 Backend Core (backend/)

- **config/SecurityConfig.java & JwtService.java:** Stateless JWT authentication (HMAC-SHA256, 24-hr token, 7-day refresh), role-based endpoint authorization, and BCrypt encryption.
- **client/ai/AiServiceClient.java:** HTTP client querying Python AI service with an intelligent fallback estimation algorithm if ML service is unreachable.
- **client/mandi/MandiApiClient.java:** Connects to Agmarknet/Data.gov.in mandi arrival price feeds.
- **service/RecommendationService.java:** Mandi optimization engine calculating distance, modal prices, freight hauling, 2% APMC tax, and handling charges.
- **service/BuyerMatchingService.java:** Weighted matchmaking algorithm linking institutional buyers and farmers based on crop, quantity, distance, price, and quality.
- **service/TransportCostService.java & HaversineUtil.java:** Great-circle distance calculations and payload tonnage logistics cost estimation.
- **service/MarketPriceSyncService.java:** Automated cron synchronization executing daily at 6:00 AM.
- **service/OfferService.java & ProfitCalculatorService.java:** Digital trade contract lifecycle management and comparative margin analysis.
- **controller/*.java:** 9 REST Controllers (Auth, Farmer, Buyer, Market, Matching, Offer, Prediction, Recommendation, Admin).
- **db/migration/V1 & V2:** Flyway SQL schemas for 10 relational tables (users, profiles, crops, varieties, markets, prices, requirements, offers) and master seed data.

C. React 19 Frontend (frontend/)

- **src/services/locationService.js:** HTML5 browser GPS integration with OpenStreetMap Nominatim reverse geocoding.
- **src/services/centralDatabase.js:** Local file-synchronized reactive database operating on physical JSON stores (users.json, crops.json, offers.json, trades.json, audits.json).
- **vite.config.js:** Custom Vite jsonWriterPlugin middleware for seamless disk persistence.
- **src/components/dashboard/LiveRouteMap.jsx:** Interactive route display embedding Google Maps iframe and generating direct Google Maps driving navigation.
- **src/pages/farmer/*.jsx:** Farmer Dashboard, Market Recommendations, Live Route Map, Price Prediction (Recharts), Profit Calculator, Buyer Marketplace, Crop Listings, and Trade Negotiations.
- **src/pages/buyer/*.jsx & mandi/*.jsx:** Buyer Procurement Dashboard, AI Matches, Inbound Offers, Mandi Directory, Real-Time Prices, and Trend Analytics.

4. Core Mathematical Formulas & Scoring Algorithms

Algorithm / Formula	Mathematical Definition	Weight & Logic Breakdown
Mandi Net Return Formula	$\text{Net} = \text{Gross} - \text{Transport} - \text{Handling} - \text{Commission}$ $\text{where Commission} = \text{Gross} * 0.02, \text{ Handling} = \text{Rs. } 30/\text{quintal}$	Calculates true cash take-home in farmer's pocket after deducting all logistics and yard fees.
Mandi Recommendation Score	$\text{Score} = (0.40 * \text{Price}) + (0.35 * \text{NetReturn}) + (0.10 * \text{Distance}) + (0.10 * \text{Trend}) + (0.05 * \text{Demand})$	Multi-criteria scoring model ranking nearest profitable APMC markets.
Buyer-Farmer Match Score	$\text{Score} = (0.30 * \text{Crop}) + (0.20 * \text{Qty}) + (0.15 * \text{Dist}) + (0.20 * \text{Price}) + (0.10 * \text{Date}) + (0.05 * \text{Quality})$	Evaluates mutual compatibility between institutional buyers and harvesting farmers.
Freight Transport Cost	$\text{Cost} = \text{BaseCost} + (\text{DistanceKm} * \text{RatePerKm} * \text{LoadFactor})$ $\text{Base} = \text{Rs. } 300, \text{ Rate} = \text{Rs. } 20/\text{km}, \text{ LoadFactor} = \max(1.0, \text{Quintals}/10)$	Logistics cost estimation model factoring hauling distance and payload tonnage.
Haversine Distance Formula	$a = \sin^2(d\text{Lat}/2) + \cos(\text{lat1}) * \cos(\text{lat2}) * \sin^2(d\text{Lon}/2)$ $d = 2 * R * \text{atan2}(\sqrt{a}, \sqrt{1-a}),$ $\text{where } R = 6371 \text{ km}$	Calculates accurate great-circle ground distance between GPS coordinates.

5. Step-by-Step Operational Workflows

Workflow 1: Farmer Discovery & Mandi Navigation Flow

- Farmer logs in, inputs harvest details (e.g. 20 Quintals of Onion), and clicks 'Detect GPS Location'.
- Browser Geolocation captures coordinates -> locationService.js queries OpenStreetMap Nominatim -> identifies village/district.
- Backend RecommendationService.java calls Python AI forecasting, computes post-freight net margins across Mandis, and assigns Gold/Silver/Bronze badges.
- Farmer opens LiveRouteMap.jsx, inspects live route corridor, and clicks 'Launch Google Maps Navigation' for turn-by-turn truck driving navigation.

Workflow 2: Direct Buyer Procurement & Digital Contract Negotiation

- Institutional Buyer publishes demand specification (e.g. 100 Quintals Tomato @ Rs. 3,400/q).
- BuyerMatchingService.java matches regional farmers with matching harvest lots.
- Farmer sends trade offer; buyer can accept or submit counter-bid via the offer state machine.
- Upon acceptance, the trade agreement is locked and appended to the immutable audit trail (audits.json / PostgreSQL).